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*Ascidia*



*Salpa*



*Doliolum*

**Figure 2.21** Examples of Urochordata

the pharynx. Notochord is present only in the tail region of the larval stage, hence named urochordata. Alimentary canal is complete and circulatory system is of open type. The heart is ventral and tubular. Respiration is through gill slits and clefts. Dorsal tubular nerve cord is present only in the larval stage and a single dorsal ganglion is present in the adults. Mostly hermaphrodites, development indirect and includes a free swimming tadpole larva with chordate characters. Retrogressive metamorphosis is seen (Figure 2.21).

Examples: *Ascidia, Salpa, Doliolum*

## 2.4.2. Subphylum: Cephalochordata

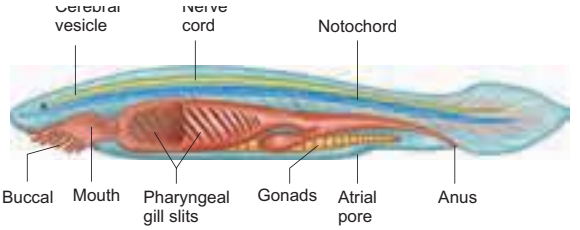
( L. *Cephalo-* 'head' ; G. *chorda* 'cord'.*)*

Cephalochordates are marine forms, found in shallow waters, leading a burrowing mode of life. They are small fish like coelomate forms with chordate characters such us notochord, dorsal tubular nerve cord and pharyngeal gill slits throughout their life. Closed type of circulatory system is seen without heart. Excretion is by protonephridia. Sexes are separate, Fertilization is external. Development is indirect and includes a free swimming larva (Figure 2.22).

Example: *Branchiostoma* (Amphioxus or lancelet)



*Amphioxus*




*Internal sturcutre of Amphioxus*

**Figure 2.22** Example of Cephalochordata

## 2.4.3. Subphylum-Vertebrata

(L. *Vertebrus* –back bone )

Vertebrates are also called higher chordates and they possess notochord during embryonic stage only. The notochord is replaced by a cartilaginous or bony vertebral column in the adult. **Hence all vertebrates are chordates but all chordates are not vertebrates.** Vertebrates possess paired appendages such as fins or limbs. Skin is covered by protective skeleton comprising of scales, feathers, hairs, claws, nails,